5.3.3 Hashes in Ethereum block header

Recall from chapter 1 that the blockchain is a tamperproof immutable ledger consisting of blocks that contain records of transactions, mutable state, logs, return values (receipts), and many other details, as shown in the Ethereum block header diagrams in figure 5.10. Txs, state, logs, and receipts are stored in a Merkle tree (trie) data structure, and the hash of this tree is stored in the header. The header also stores a hash of the previous block's header, forming a link to the previous block, constructing the chain, and enforcing immutability. Even a single bit change in the block's contents will change its hash significantly, thus breaking the chain; so as you can see, the block hashes are instrumental in realizing the immutability and integrity of the chain!

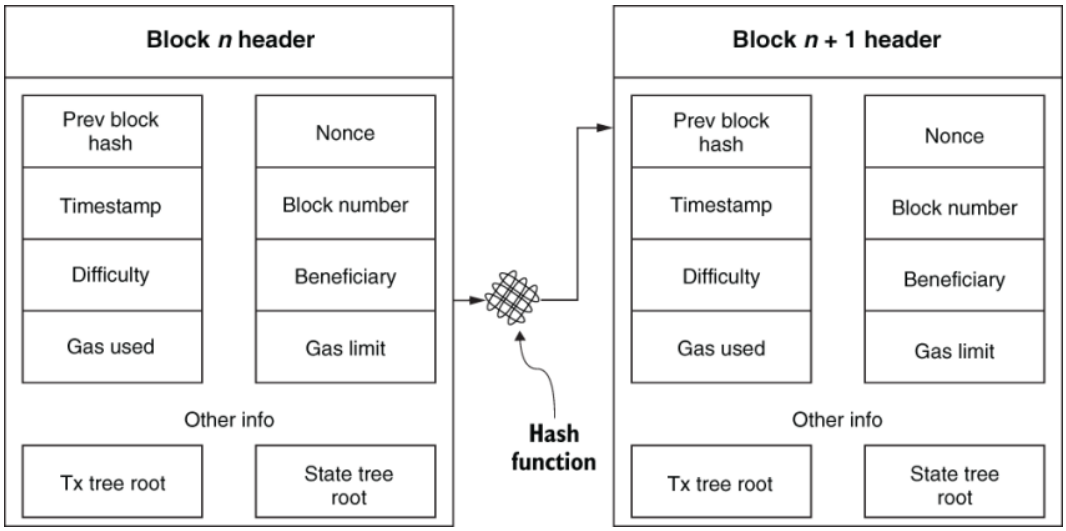


Figure 5.10 Ethereum block headers (partial) for block n and block n+1

Hashing is a core component of the consensus process for deciding the next block to be appended to the chain. Hashing is also a recommended preprocessing step for encryption of messages and digital signing of transactions. It plays an important role not only at the protocol level, but also at the application level.